Infrared Spectroscopy

1. Which of the following molecules are IR active? (10 pts)
   (a) N_2, (b) HBr, (c) CH_4, (d) Xe, (e) H_2O_2, (f) NO

2. Draw a vibrational mode of the BF_3 molecule that is IR inactive. (10 pts)

3. Calculate the number of vibrational degrees of freedom of the hemoglobin molecule, which contains 9272 atoms. (10 pts)

4. The typical energy difference for IR is 0.5×10^{-19} J. Calculate the ratio of the number of molecules in two adjacent energy levels (for example, the ground level and the first excited level) at 300 K. (10 pts)

5. A molecule XY_2 is known to be linear, but it is not clear whether it is Y-X-Y or X-Y-Y. How would you use IR spectroscopy to determine its structure? (20 pts)

6. The fundamental frequency of vibration for carbon monoxide (^{12}CO) is 2143.3 cm^{-1}. Calculate the force constant for the carbon-oxygen bond. If ^{12}C is replaced by ^{13}C, predict the fundamental frequency of vibration for ^{13}CO. (40 pts)